

$$\begin{aligned} & \frac{1}{16\pi^2} \left( -\frac{1}{144} \frac{1}{m_e^2} s_\gamma^2 (108 c_\gamma^2 \overline{y_e^{r13}} \overline{y_e^{i2p}} y_e^{r14} y_e^{i1p} + 27 \overline{y_e^{pr}} \overline{y_e^{i2i3}} y_e^{p14} y_e^{i1r} (1 + s_\gamma^2) - \right. \\ & \quad 72 c_\gamma^2 \overline{y_e^{pr}} \overline{y_e^{i2i3}} y_e^{pr} y_e^{i1i4} + 9 y_e^{i1i4} (\overline{y_e^{i2i3}} (38 g_1^2 + 6 g_2^2) + 3 \overline{y_e^{p13}} \overline{y_e^{i2r}} y_e^{pr}) + \\ & \quad \left. \overline{y_e^{p13}} (27 s_\gamma^2 \overline{y_e^{i2r}} y_e^{pr} y_e^{i1i4} + 10 g_1^2 y_e^{p14} \delta_{i1i2}) + 14 g_1^2 \overline{y_e^{i2p}} y_e^{i1p} \delta_{i3i4} \right) - \\ & \frac{1}{4} \sum_p s_\gamma g_1^2 \frac{1}{m_e^4} \overline{y_e^{i2i3}} y_e^{i1i4} (2 s_{2\gamma} c_\gamma + s_\gamma c_{2\gamma}) \text{LF}_{1,0}[m_d^p] + \frac{27}{27} \sum_p g_1^4 \text{LF}_{3,0}[m_d^p] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{5}{36} \sum_p g_1^4 \text{LF}_{4,-1}[m_d^p] \delta_{i1i2} \delta_{i3i4} + \frac{8}{135} \sum_p g_1^4 \text{LF}_{5,-2}[m_d^p] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{3}{2} s_\gamma \frac{1}{m_e^4} \overline{y_d^{pr}} y_d^{pr} \overline{y_e^{i2i3}} y_e^{i1i4} (-s_{2\gamma} c_\gamma + s_\gamma^3) \text{LF}_{1,0}[m_d^r] - \\ & \frac{1}{4} \sum_p s_\gamma g_1^2 \frac{1}{m_e^4} \overline{y_e^{i2i3}} y_e^{i1i4} (2 s_{2\gamma} c_\gamma + s_\gamma c_{2\gamma}) \text{LF}_{1,0}[m_e^p] + \frac{2}{9} \sum_p g_1^4 \text{LF}_{3,0}[m_e^p] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{5}{12} \sum_p g_1^4 \text{LF}_{4,-1}[m_e^p] \delta_{i1i2} \delta_{i3i4} + \frac{8}{45} \sum_p g_1^4 \text{LF}_{5,-2}[m_e^p] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{1}{2} s_\gamma \frac{1}{m_e^4} \overline{y_e^{pr}} \overline{y_e^{i2i3}} y_e^{pr} y_e^{i1i4} (-s_{2\gamma} c_\gamma + s_\gamma^3) \text{LF}_{1,0}[m_e^r] + \\ & \frac{1}{4} s_\gamma \frac{1}{m_e^4} \overline{y_e^{i2i3}} y_e^{i1i4} (2 \overline{y_e^{pr}} y_e^{pr} (s_{2\gamma} c_\gamma - s_\gamma^3) + \sum_p g_1^2 (2 s_{2\gamma} c_\gamma + s_\gamma c_{2\gamma})) \text{LF}_{1,0}[m_l^p] + \\ & \frac{1}{9} \sum_p g_1^4 \text{LF}_{3,0}[m_l^p] \delta_{i1i2} \delta_{i3i4} - \frac{5}{24} \sum_p g_1^4 \text{LF}_{4,-1}[m_l^p] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{4}{45} \sum_p g_1^4 \text{LF}_{5,-2}[m_l^p] \delta_{i1i2} \delta_{i3i4} - \frac{1}{4} s_\gamma \frac{1}{m_e^4} \overline{y_e^{i2i3}} y_e^{i1i4} \\ & \quad (6 \overline{y_d^{pr}} y_d^{pr} (-s_{2\gamma} c_\gamma + s_\gamma^3) + 6 c_\gamma \overline{y_u^{pr}} y_u^{pr} (s_{2\gamma} + s_\gamma c_\gamma) + \sum_p g_1^2 (2 s_{2\gamma} c_\gamma + s_\gamma c_{2\gamma})) \\ & \quad \text{LF}_{1,0}[m_q^p] + \frac{1}{27} \sum_p g_1^4 \text{LF}_{3,0}[m_q^p] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{5}{72} \sum_p g_1^4 \text{LF}_{4,-1}[m_q^p] \delta_{i1i2} \delta_{i3i4} + \frac{4}{135} \sum_p g_1^4 \text{LF}_{5,-2}[m_q^p] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{2} \sum_p s_\gamma g_1^2 \frac{1}{m_e^4} \overline{y_e^{i2i3}} y_e^{i1i4} (2 s_{2\gamma} c_\gamma + s_\gamma c_{2\gamma}) \text{LF}_{1,0}[m_u^p] + \frac{8}{27} \sum_p g_1^4 \text{LF}_{3,0}[m_u^p] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{5}{9} \sum_p g_1^4 \text{LF}_{4,-1}[m_u^p] \delta_{i1i2} \delta_{i3i4} + \frac{32}{135} \sum_p g_1^4 \text{LF}_{5,-2}[m_u^p] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{3}{2} s_\gamma c_\gamma \frac{1}{m_e^4} \overline{y_e^{i2i3}} y_e^{i1i4} \overline{y_u^{pr}} y_u^{pr} (s_{2\gamma} + s_\gamma c_\gamma) \text{LF}_{1,0}[m_u^r] - \frac{1}{8} s_\gamma \frac{1}{m_e^4} \overline{y_e^{i2i3}} y_e^{i1i4} \\ & \quad (3 s_{4\gamma} c_\gamma (g_1^2 + g_2^2) + s_\gamma (g_1^2 (-1 + 3 c_{2\gamma}^2) + 3 g_2^2 (-1 + c_{2\gamma}^2))) \text{LF}_{1,0}[m_\oplus] + \\ & \frac{1}{8} \frac{1}{m_e^2} (3 s_\gamma^4 \overline{y_e^{pr}} \overline{y_e^{i2i3}} y_e^{p14} y_e^{i1r} + s_\gamma^2 y_e^{i1i4} (-2 \overline{y_e^{i2i3}} (g_1^2 + 3 g_2^2) + 3 s_\gamma^2 \overline{y_e^{p13}} \overline{y_e^{i2r}} y_e^{pr})) \\ & \quad \text{LF}_{1,1}[m_\oplus] + \frac{1}{24} s_\gamma^2 (3 \overline{y_e^{i2i3}} y_e^{i1i4} (g_1^2 + 3 g_2^2) - 4 g_1^2 \overline{y_e^{p13}} y_e^{p14} \delta_{i1i2} - \\ & \quad 4 \overline{y_e^{i2p}} y_e^{i1p} (3 c_\gamma^2 \overline{y_e^{r13}} y_e^{r14} + 2 g_1^2 \delta_{i3i4})) \text{LF}_{1,2}[m_\oplus] + \\ & \frac{1}{12} s_\gamma^2 (-g_1^2 \overline{y_e^{p13}} y_e^{p14} \delta_{i1i2} + \overline{y_e^{i2p}} y_e^{i1p} (-3 s_\gamma^2 \overline{y_e^{r13}} y_e^{r14} + g_1^2 \delta_{i3i4})) \text{LF}_{2,1}[m_\oplus] + \\ & \frac{1}{36} g_1^2 (3 s_\gamma^2 \overline{y_e^{p13}} y_e^{p14} \delta_{i1i2} + (-3 s_\gamma^2 \overline{y_e^{i2p}} y_e^{i1p} + 4 g_1^2 \delta_{i1i2}) \delta_{i3i4}) \text{LF}_{3,0}[m_\oplus] - \\ & \frac{5}{24} g_1^4 \text{LF}_{4,-1}[m_\oplus] \delta_{i1i2} \delta_{i3i4} + \frac{4}{45} g_1^4 \text{LF}_{5,-2}[m_\oplus] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{9} g_1^4 \text{LF}_{3,0}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{6} g_1^4 \text{LF}_{4,-1}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{8}{45} g_1^4 \text{LF}_{5,-2}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{2} g_1^2 \frac{1}{m_e^2} s_\gamma^2 \overline{y_e^{i2i3}} y_e^{i1i4} \text{LF}_{1,1,0}[m_1, m_e^{i3}] - \\ & \frac{1}{4} g_1^2 \frac{1}{m_e^2} s_\gamma^2 \overline{y_e^{i2i3}} y_e^{i1i4} \text{LF}_{2,1,-1}[m_1, m_e^{i3}] + \frac{1}{12} g_1^4 \text{LF}_{2,1,0}[m_1, m_e^{i3}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{12} g_1^4 \text{LF}_{2,2,-1}[m_1, m_e^{i3}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{6} g_1^4 \text{LF}_{3,1,-1}[m_1, m_e^{i3}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{12} g_1^4 \text{LF}_{4,1,-2}[m_1, m_e^{i3}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{2} g_1^2 \frac{1}{m_e^2} s_\gamma^2 \overline{y_e^{i2i3}} y_e^{i1i4} \text{LF}_{1,1,0}[m_1, m_e^{i4}] - \\ & \frac{1}{4} g_1^2 \frac{1}{m_e^2} s_\gamma^2 \overline{y_e^{i2i3}} y_e^{i1i4} \text{LF}_{2,1,-1}[m_1, m_e^{i4}] + \frac{1}{12} g_1^4 \text{LF}_{2,1,0}[m_1, m_e^{i4}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{12} g_1^4 \text{LF}_{2,2,-1}[m_1, m_e^{i4}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{6} g_1^4 \text{LF}_{3,1,-1}[m_1, m_e^{i4}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{12} g_1^4 \text{LF}_{4,1,-2}[m_1, m_e^{i4}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{8} g_1^2 \frac{1}{m_e^2} s_\gamma^2 \overline{y_e^{i2i3}} y_e^{i1i4} \text{LF}_{1,1,0}[m_1, m_l^{i1}] - \\ & \frac{1}{16} g_1^2 \frac{1}{m_e^2} s_\gamma^2 \overline{y_e^{i2i3}} y_e^{i1i4} \text{LF}_{2,1,-1}[m_1, m_l^{i1}] + \frac{1}{48} g_1^4 \text{LF}_{2,1,0}[m_1, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{48} g_1^4 \text{LF}_{2,2,-1}[m_1, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{24} g_1^4 \text{LF}_{3,1,-1}[m_1, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{48} g_1^4 \text{LF}_{4,1,-2}[m_1, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{8} g_1^2 \frac{1}{m_e^2} s_\gamma^2 \overline{y_e^{i2i3}} y_e^{i1i4} \text{LF}_{1,1,0}[m_1, m_l^{i2}] - \\ & \frac{1}{16} g_1^2 \frac{1}{m_e^2} s_\gamma^2 \overline{y_e^{i2i3}} y_e^{i1i4} \text{LF}_{2,1,-1}[m_1, m_l^{i2}] + \frac{1}{48} g_1^4 \text{LF}_{2,1,0}[m_1$$